Sohum Mehta

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Education

University of Pennsylvania — Philadelphia, PA
Cognitive Science & Computer Science (BA); Minor: Statistics and Data Science (Wharton) — *May 2030*

Illinois Mathematics and Science Academy — Aurora, IL
High School Diploma | GPA: 3.98/4.00 (Unweighted)
- **Relevant Coursework**: Machine Learning, OOP, Multi-variable Calculus, Differential Equations, NeuroAI

Y Combinator Startup School '26 | *Became a Founder in San Francisco + research presentations & hackathons*

Experience

Google Quantum AI | *Explainable AI Research Intern* — *06/2025 – 07/2026*
- Research variational quantum algorithms (**VQE/VQC**) and explainable AI under Software Lead mentorship
- Built reproducible hybrid **QML** pipelines; benchmarks informed internal research directions
- Accepted at **IEEE/ACM AI-ML Systems** (Quantum Track), **ML4PS @ NeurIPS 2025**, and **YQIS 2025**

Wedge (YC S25) | *Internal Project Fellow, Software Engineering & Research* — *08/2025*
- Only high-school operator selected for this **YC-backed fellowship (<0.7% chosen)**, with undergrad/grad teams
- Led applied-AI and workflow-automation work in Edge Consulting for **major hospital systems**
- Bridged prototypes to client-ready solutions with SWE/research teams; peers billed up to **$180/hr**

Winter Lab, Northwestern (Feinberg) | *Bioinformatics Research Intern* — *06/2025 – 09/2025*
- Engineered a reusable **single-cell RNA-seq web app** for lab-wide analysis of immune-cell gene programs
- Led biomarker discovery across inflammatory disease datasets; immunologists validated signals
- First-author on the resulting **ACM-BCB 2025** proceedings paper (poster presentation)

Carnegie Mellon University | *Computational Neurolinguistics Student Researcher* — *09/2024 – 05/2025*
- Modeled lexical retrieval from **EEG** and reaction-time data (∼20,000 points) across picture vs. word processing
- **National JSHS Finalist** (top ∼3%, $1,000); oral presentation accepted at **Springer Brain Informatics**
- Built Python notebooks and regression workflows that generalized across datasets and stimuli types

Selected Publications and Presentations

TabFaith: Benchmarking and Improving Structural Faithfulness in LLM Table Summarization — ACL 2026 SURGeLLM (Oral)

QCxAI: Parameter-Shift Saliency for Variational Quantum Classifiers — NeurIPS 2025 ML4PS

Prompt Genotyping: Quantifying the Evaluation Gap Between Synthetic Benchmarks and Real LLM Performance — EMNLP 2025 BlackBox

Cryptographic Fingerprinting for Medical AI — NeurIPS 2025 Lock-LLM

Cell-Type-Aware Single-Cell ML Reveals Biomarkers for Sarcoidosis Prognosis — ACM-BCB 2025

Projects & Activities

Promptonium | *Next.js, TypeScript, LLM APIs, Evaluation Pipelines, Vercel*
- Research-backed platform for effective prompt curation – termed **prompt genotyping**
- **Microsoft and OpenAI-supported ($2K)**; research framework cited by Carnegie Mellon and Microsoft
- Shipped an educational prompting tool on the framework; used by PhD researchers

TabFaith | *Python, LLM Evaluation, NLI, ToTTo, Data Annotation*
- Co-created a **2,400-example benchmark** for factual errors in LLM-generated table summaries
- Co-developed a **structure-aware faithfulness metric** and training-free pipeline to identify and repair errors
- Published at **ACL SURGeLLM 2026**; selected as a **top-six oral paper from over 90 submissions**

BiteSense | *Python, TensorFlow/Keras, ResNet50, Flask, JavaScript*
- Trained an **8-class ResNet50** to **81.25%** accuracy from a **43.75%** baseline across 1,925 images
- Fine-tuned the top **10 layers** at **1e-4** after frozen-backbone transfer with 6-way augmentation
- Won **Rep. LaHood's Congressional App Challenge (IL-16)**: 1 of 382 winners from **3,881 apps**

Pay it Forward (PiF) Science | *Co-Founder & Co-President, 501(c)(3) Nonprofit* — *Ongoing*
- Democratized STEM access for **3,000+ students** through weekly workshops and learning tools
- Recruited and led ∼**70 advisors**; built **9 international NGO partnerships** to expand programming
- Sustained by a **Google Ads Grant (∼$120K/yr)**, scaling multi-site delivery and campaigns

Technical Skills

Languages: Python, Java, C#, TypeScript, MATLAB, R, HTML/CSS, JavaScript
Frameworks: PyTorch, TensorFlow/Keras, scikit-learn, Hugging Face, Flask, React, Next.js, Cirq
Developer Tools: Git, LaTeX, Vercel, REST APIs, Jupyter, Microsoft Office 365
Computational Concepts: Explainable AI, NLP, Quantum ML, Computational Neuroscience, Bioinformatics